

# Isais Marrufo

Los Angeles, CA | isaismarrufo@gmail.com | (530)-854-9682 | linkedin.com/in/isaismarrufo | github.com/isaismarrufo

## EDUCATION

**University of Southern California** | BS Computer Engineering & Computer Science

**Graduation Date: May 2027**

GPA: 3.6/4.0

## EXPERIENCE

### **Butte College**

*August 2022 – August 2025*

*School Certifying Official – Assistant*

*Oroville, CA*

- Managed and certified veteran student enrollments for 150+ students, ensuring compliance with VA guidelines
- Maintained and updated veteran student records using Enrollment Manager software
- Guided veteran students through the class registration process, course requirements, and schedules
- Developed and updated procedural manuals for veterans using Colleague software

### **United States Army - Various Locations**

*January 2018 – June 2022*

*Supervisor, Vehicle Recovery/Repair*

*Fort Riley, KS*

- Led a team of 8 mechanics and oversaw service and recovery operations on 51 heavy tracked and light vehicles
- Provided technical guidance and mentored team members on proper troubleshooting procedures in compliance with Technical Manual 9-2350-292, researched Class IX repairable parts, and coordinated timely parts requisition
- Filled mission critical vacant supervisor role, increased, and maintained operational readiness of 4 Recovery Vehicles
- Accounted for 5 General Mechanics Tool Kits and 2 M88A2 Heavy Recovery Vehicles valued at \$3M+
- Mentored 4 crews on recovery operations during 3 large-scale training exercises, and completed 50+ recoveries
- Supervised and trained a team of 6 in rigging techniques and expedient repairs, proper towing, lifting, winching, and accounted for \$2M worth of equipment
- Supported junior advancement by coordinating 60,000 cumulative hours of military education, leading to 3 promotions

## PROJECTS

### **FPGA-Based Interactive System (Nexys A7)**

*March 2026 – May 2026*

*Digital Design*

- Designed a synchronous RTL system in Verilog using a multi-state finite state machine (FSM) to control sequence generation, input validation, and state transitions
- Developed debounced button input and one-pulse logic to ensure deterministic user input under hardware timing constraints
- Implemented a Linear-Feedback Shift Register (LFSR)-based pseudo-random sequence generator using XOR feedback for hardware-efficient pattern generation
- Built and verified a VGA timing and rendering module (640x480), debugging synchronization and timing issues via simulation and on-board testing

### **Dual-Sensor Speed Trap System**

*October 2025 – December 2025*

*Embedded Systems*

- Developed embedded C firmware on an AVR microcontroller using timers, interrupts, ADC, GPIO, EEPROM, and PWM to perform real-time speed measurements with optical sensors
- Implemented low-level device drivers for LCD, rotary encoder, servo motor, RGB LED, and buzzer using direct register access and fixed-point arithmetic under real-time constraints
- Built, tested, and debugged a multi-file embedded project with Makefiles, producing ELF/HEX binaries and validating timing accuracy and I/O behavior on physical hardware

## SKILLS

- Languages: C, C++, Python, Go, x86 Assembly, PicoBlaze
- Design & Simulation Tools: Xilinx Vivado, QuestaSim, GTKWave
- Hardware Description Languages: Verilog, VHDL
- Systems & Development Tools: Linux, CLI, Git, GitHub, Docker, Telegraf, InfluxDB, Grafana
- Communication Protocols: MQTT, TCP/IP, HTTP, GPIO, I2C, SPI, UART